

HARDWARE DESIGN SPECIFICATION

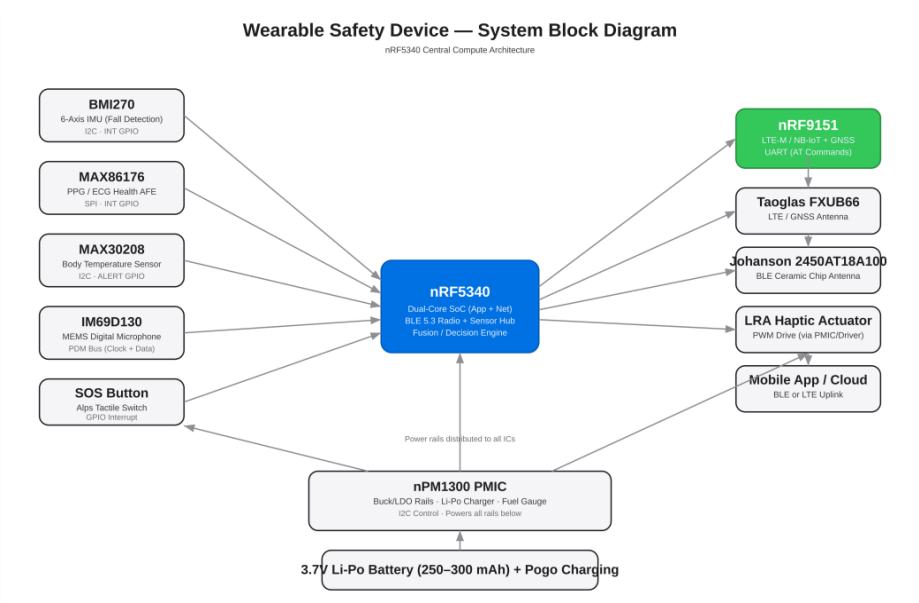
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AI-Powered Smart Wrist Safety Wearable

Continuous Health Monitoring · Fall & Emergency Detection · Autonomous LTE Alerting

Document Class:

Pre-Schematic Hardware Architecture & Component Specification



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3. Hardware System Overview

This document defines the complete hardware architecture for a premium AI-powered wrist safety wearable. Unlike a conventional smartwatch, this device has no display — its purpose is silent, continuous background monitoring: motion, cardiac signal, body temperature, and ambient sound are all sampled continuously and analyzed on-device. When the onboard AI detects an abnormal pattern (a fall, an irregular heart rate, an unusual temperature, or a distress sound), the device decides autonomously how to respond: a haptic vibration, a Bluetooth notification to the paired phone, or — if no phone is nearby — a direct LTE-M/NB-IoT emergency transmission with GNSS location.

Every hardware decision in this document flows from one architectural principle: the wearable must never depend entirely on a paired phone to save its wearer. That single requirement is why the design includes a full cellular modem (nRF9151) in addition to Bluetooth, and why the emergency button exists as a manual, AI-independent fallback.

1.1 High-Level Architecture

At the center of the system sits the nRF5340, a dual-core Bluetooth SoC that acts as the "brain": it reads every sensor, runs the on-device decision logic, and either talks to the phone over BLE or hands the emergency payload to the nRF9151 cellular modem. Power for the whole system is managed by a single PMIC (nPM1300), which charges the battery, produces every voltage rail, and reports battery state back to the MCU.

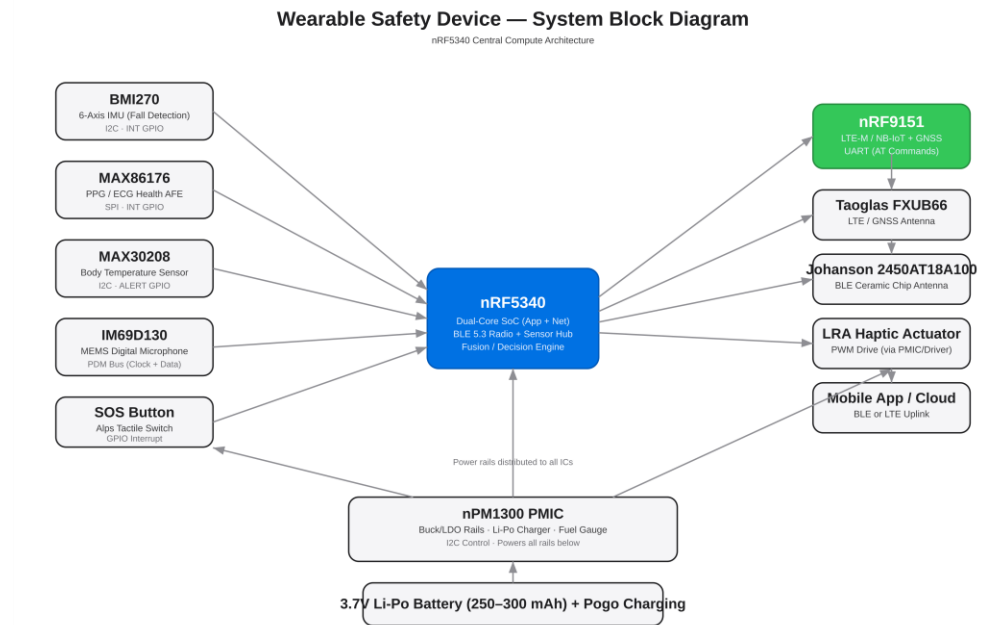


Figure 1.1 — Full system block diagram: sensors, compute, connectivity, and power.

1.2 Component Communication Summary

- Sensors (BMI270, MAX86176, MAX30208, IM69D130) communicate to the nRF5340 over I2C, SPI, or PDM — see Section 10 for the full interface map.
- The nRF5340 communicates with the nRF9151 cellular modem over UART using AT commands, the same way a computer talks to an external modem.

- The nPM1300 PMIC communicates its status (battery %, charge state, faults) to the nRF5340 over I2C, and receives configuration commands the same way.
- The haptic actuator is driven by a dedicated haptic driver IC over PWM/I2C — the MCU never drives the LRA directly.
- Both antennas are purely passive RF elements — they do not "communicate" digitally; they radiate whatever signal the nRF5340 (BLE) or nRF9151 (LTE/GNSS) feeds them.

Note: This document treats all component values (resistors, capacitors, crystal loads, matching-network components) as reference-design starting points. Every value must be re-verified against the latest datasheet/reference design revision during schematic capture, and RF matching networks must be tuned against the real PCB and enclosure using a network analyzer before production.

4. Complete Hardware Component List

The table below is the master reference for every major hardware component in the design. It intentionally repeats some information found later in this document (Section 5, Section 14) so that it can be used standalone as a quick-reference sheet during schematic capture.

nRF5340 SoC

Field	Detail
Manufacturer	Nordic Semiconductor
Part Number	nRF5340-QKAA-R
Category	Main MCU / Application Processor
Function	Central compute: runs sensor fusion, on-device AI/ML inference, power orchestration, and the Bluetooth LE stack.
Why Selected	Only MCU on the market combining a dedicated Cortex-M33 Application core (128 MHz) with a separate Cortex-M33 Network core (64 MHz) for BLE — lets AI inference run without ever blocking the radio, at very low power.
Package	AQFN73, 7×7 mm, 0.5 mm pitch
Dimensions	7 × 7 × 0.85 mm
Voltage	1.7V–5.5V (internal regulators generate 1.8V/3.0V rails)
Power Consumption	Active RX ~4.6 mA, TX(0dBm) ~4.9 mA, System-OFF ~0.4 µA
Communication Interface	I2C, SPI, UART, PDM, I2S, GPIO, QSPI, BLE 5.3 radio
Typical Applications	Wearables, medical sensors, asset trackers, smart-home BLE nodes
Advantages	Dual-core isolation, TrustZone security, mature SDK (nRF Connect), tiny package
Limitations	Requires careful RF layout; app-core RAM (512KB) can be tight for heavy on-device AI models
Estimated Price	\$6.50–\$8.00 (1k units)
Datasheet Reference	nordicsemi.com — nRF5340 Product Specification v1.5

nRF9151 SiP

Field	Detail
Manufacturer	Nordic Semiconductor
Part Number	nRF9151-LACA-R
Category	Cellular Connectivity (LTE-M / NB-IoT + GNSS)
Function	Provides autonomous, phone-independent emergency communication over cellular LTE-M/NB-IoT, plus GNSS positioning for location-tagged alerts.
Why Selected	Smallest and lowest-power LTE-M/NB-IoT+GNSS SiP available; integrates RF front-end, power management, and application MCU in one part, minimizing PCB area — critical in a slim wrist wearable.

Field	Detail
Package	LGA, System-in-Package
Dimensions	≈6.9 × 8.98 × 1.2 mm
Voltage	3.0V–4.4V direct from battery-adjacent rail (integrated PMIC inside SiP)
Power Consumption	Idle (PSM) <3 μA, Active RX ~50 mA, TX peak up to ~500 mA (short bursts)
Communication Interface	UART (AT command set), SPI, I2C, GPIO
Typical Applications	Asset trackers, medical alert devices, industrial IoT, smart utility meters
Advantages	Autonomous cellular link independent of a paired phone; built-in GNSS; global multi-band support
Limitations	Highest peak current draw in the system — battery and PMIC rail must be sized for LTE TX bursts; requires a nano-SIM or eSIM profile
Estimated Price	\$12–\$15 (1k units)
Datasheet Reference	nordicsemi.com — nRF9151 Product Specification v1.0

BMI270 IMU

Field	Detail
Manufacturer	Bosch Sensortec
Part Number	BMI270
Category	Motion Sensor (6-axis IMU)
Function	Detects motion, orientation, steps, and — critically — free-fall / impact signatures used for automatic fall detection.
Why Selected	Ultra-low-power (down to 685 μA) with on-chip motion/step/tap recognition, so the nRF5340 can stay asleep and only wake on a qualifying motion interrupt — essential for multi-day battery life.
Package	LGA-14
Dimensions	2.5 × 3.0 × 0.83 mm
Voltage	1.71V–3.6V
Power Consumption	685 μA (full operation), 1.8 μA (accel low-power)
Communication Interface	I2C (400kHz) or SPI (10MHz), 2× interrupt GPIO lines
Typical Applications	Wearables, drones, AR/VR, pedometers
Advantages	Tiny footprint, built-in step counter & activity classifier, low noise
Limitations	No magnetometer (no absolute heading) — acceptable since this design only needs relative motion for fall/activity detection
Estimated Price	\$2.20–\$2.80 (1k units)
Datasheet Reference	bosch-sensortec.com — BMI270 Datasheet Rev. 1.9

MAX86176 Health AFE

Field	Detail
Manufacturer	Analog Devices (Maxim)
Part Number	MAX86176
Category	Health Sensor (PPG / ECG Analog Front-End)
Function	Optical PPG for heart rate & SpO ₂ , plus single-lead ECG capability — the core of continuous cardiac/health monitoring.
Why Selected	Integrates the LED drivers, photodiode transimpedance amplifier, and ECG analog front-end in one part with a high dynamic-range ADC, reducing external analog circuitry versus a discrete design.
Package	WLP / TQFN (ultra-small)
Dimensions	≈3.1 × 3.1 mm
Voltage	Analog 1.8V/3.3V, LED supply up to 5V
Power Consumption	≈1–3 mA typical during active PPG sampling
Communication Interface	SPI (up to 8+ MHz) + interrupt GPIO
Typical Applications	Fitness trackers, medical-grade wearables, patches
Advantages	High SNR optical front-end, flexible LED current control, on-chip ECG
Limitations	Optical accuracy sensitive to skin contact/motion — requires a good mechanical optical window and skin seal
Estimated Price	\$4–\$6 (1k units)
Datasheet Reference	analog.com — MAX86176 Data Sheet

MAX30208 Temp Sensor

Field	Detail
Manufacturer	Analog Devices (Maxim)
Part Number	MAX30208
Category	Body Temperature Sensor
Function	High-resolution digital skin-temperature measurement, feeding both trend logging and abnormal-temperature emergency detection.
Why Selected	±0.1°C clinical-grade accuracy in an ultra-tiny package with a simple I2C interface and a programmable ALERT threshold — lets the sensor itself flag an abnormal reading without CPU polling.
Package	UTDFN / WLP
Dimensions	1.6 × 1.6 × 0.5 mm
Voltage	1.7V–3.6V
Power Consumption	≈50 µA average (low duty-cycle sampling)
Communication Interface	I2C (up to 400kHz) + ALERT GPIO

Field	Detail
Typical Applications	Wearable thermometers, cold-chain tags, medical patches
Advantages	Extremely small, high accuracy, low power, hardware alert threshold
Limitations	Measures skin (not core) temperature — inherent physiological offset must be compensated in firmware/app
Estimated Price	\$1.50–\$2.00 (1k units)
Datasheet Reference	analog.com — MAX30208 Data Sheet

IM69D130 MEMS Microphone

Field	Detail
Manufacturer	Infineon Technologies
Part Number	IM69D130
Category	Audio Sensor (Digital MEMS Microphone)
Function	Captures ambient audio for AI-based sound event detection (e.g., distress calls, crash/impact sounds, screams) as a secondary emergency-detection channel.
Why Selected	Industry-leading 69 dB(A) SNR and 130 dB SPL acoustic overload point — a high dynamic range needed to reliably classify both quiet distress sounds and sudden loud events without clipping.
Package	Bottom-port SMD MEMS package
Dimensions	4.72 × 3.76 × 1.25 mm
Voltage	1.62V–3.6V
Power Consumption	≈600 μA active
Communication Interface	PDM (Pulse Density Modulation) — clock + data, no analog wiring needed
Typical Applications	Voice assistants, acoustic event detection, hearables
Advantages	Very high SNR, fully digital (low noise pickup), small package
Limitations	PDM decimation/filtering adds a small DSP load on the app core
Estimated Price	\$1.00–\$1.40 (1k units)
Datasheet Reference	infineon.com — IM69D130 Datasheet

nPM1300 PMIC

Field	Detail
Manufacturer	Nordic Semiconductor
Part Number	nPM1300
Category	Power Management IC
Function	Single-chip Li-Po battery charger, dual buck converters, dual LDOs, fuel gauge, and GPIO expansion — the power hub of the entire wearable.

Field	Detail
Why Selected	Purpose-built for Nordic wireless SoCs; integrates charge management (removing a separate charger IC), multiple regulated rails, and an accurate fuel gauge in one very small package, minimizing both BOM count and PCB area.
Package	WLCSP / QFN
Dimensions	≈3.0 × 3.0 mm
Voltage	VBAT input 2.6V–4.45V; outputs 2× Buck + 2× LDO configurable 1.7V–3.3V
Power Consumption	Quiescent current ≈ 1–2 μA (ship mode even lower)
Communication Interface	I2C (control/telemetry) + multiple GPIO + charge-status interrupt
Typical Applications	Any Li-Po powered Nordic wireless product
Advantages	Integrated charger+regulation+fuel gauge, tiny footprint, software-configurable rails
Limitations	Buck converter efficiency requires correct inductor selection; needs careful layout for thermal charging performance
Estimated Price	\$1.80–\$2.30 (1k units)
Datasheet Reference	nordicsemi.com — nPM1300 Product Specification

Li-Po Battery Pack

Field	Detail
Manufacturer	Cell vendor (e.g., Lishen / EEMB / Grepow — customer-specified)
Part Number	Custom 3.7V 250–300mAh flex pack
Category	Energy Storage
Function	Primary energy source for all onboard electronics and the LTE transmit bursts.
Why Selected	Lithium-polymer in a flexible pouch form factor is the only technology that fits a slim curved wristband while providing 1–2 days of runtime with an LTE radio onboard.
Package	Flexible pouch cell, custom curved outline
Dimensions	Application-specific, ≈0.5–4mm thick depending on capacity target
Voltage	3.7V nominal (3.0V–4.2V range)
Power Consumption	Capacity: 250–300 mAh (≈0.9–1.1 Wh)
Communication Interface	JST or direct-solder 2-wire (+/GND) to PMIC input
Typical Applications	Wearables, medical patches, slim IoT devices
Advantages	High energy density, thin/flexible form factor, rechargeable
Limitations	Requires a protection circuit module (PCM) and careful thermal/charge management; capacity trade-off vs. device thickness
Estimated Price	\$2.50–\$4.00 (1k units)
Datasheet Reference	Cell-vendor specific datasheet + UL1642/IEC62133 certification report

LRA Haptic Actuator

Field	Detail
Manufacturer	Precision Microdrives
Part Number	(e.g.) 8mm Linear Resonant Actuator, C-series
Category	Haptic Feedback
Function	Delivers crisp vibration alerts to the wearer for notifications, warnings, and emergency confirmations.
Why Selected	A Linear Resonant Actuator produces sharper, more distinct, and more energy-efficient vibration pulses than a traditional eccentric-rotating-mass motor, at a resonant frequency the driver can be tuned to — important for both perceived quality and battery life.
Package	Coin/cylindrical can
Dimensions	≈8–10 mm diameter, 3–4 mm thick
Voltage	Rated ≈1.5–2.0 Vrms drive (resonant AC drive, not DC)
Power Consumption	Peak ≈100–150 mA during active pulse, 0 mA idle
Communication Interface	Analog AC drive from a haptic driver IC (PWM-synthesized sine or dedicated haptic driver)
Typical Applications	Wearables, phones, controllers
Advantages	Fast rise/fall time, crisp tactile feel, efficient at resonance
Limitations	Must be driven at its specific resonant frequency for best performance — requires a proper haptic driver IC, not a raw GPIO/PWM pin
Estimated Price	\$1.20–\$1.80 (1k units)
Datasheet Reference	precisionmicrodrives.com — LRA series datasheet

BLE Ceramic Chip Antenna

Field	Detail
Manufacturer	Johanson Technology
Part Number	2450AT18A100
Category	RF — Bluetooth Antenna
Function	Radiates/receives the 2.4 GHz BLE signal for phone pairing and local data sync.
Why Selected	A ceramic chip antenna gives predictable, pre-characterized 2.4GHz performance in a very small SMD footprint — appropriate for a PCB too small to fit an efficient trace antenna.
Package	1.6 × 0.8 mm ceramic SMD chip
Dimensions	1.6 × 0.8 × 0.55 mm
Voltage	N/A (passive)
Power Consumption	N/A (passive, insertion loss only)
Communication Interface	RF feed line (50Ω), fed from nRF5340 ANT pin via matching network
Typical Applications	Any 2.4GHz ISM-band product (BLE/Wi-Fi/Zigbee)

Field	Detail
Advantages	Tiny size, good efficiency for its size, low cost
Limitations	Requires a clean ground-plane keep-out area and careful matching-network tuning for the final enclosure/PCB stack-up
Estimated Price	\$0.30–\$0.50 (1k units)
Datasheet Reference	johansontechnology.com — 2450AT18A100 Datasheet

LTE/GNSS Flexible Antenna

Field	Detail
Manufacturer	Taoglas
Part Number	FXUB66
Category	RF — Cellular & GNSS Antenna
Function	Wideband flexible antenna covering LTE-M/NB-IoT cellular bands plus GNSS (GPS/GLONASS/Galileo/BeiDou) reception for location.
Why Selected	A flexible adhesive-backed antenna can be routed along the curved inner wall of a small wearable case — impossible for a rigid patch or chip antenna at these lower, longer-wavelength bands — while covering the wide multi-band range LTE-M/NB-IoT/GNSS requires.
Package	Flexible PCB (FPC) antenna, adhesive-backed
Dimensions	≈70–100 mm length (folded/routed inside case), <0.3mm thick
Voltage	N/A (passive)
Power Consumption	N/A (passive)
Communication Interface	U.FL/coax pigtail or direct solder feed to nRF9151 RF pin
Typical Applications	Cellular IoT trackers, wearables, telematics
Advantages	Wide multi-band coverage, flexible mounting, combined cellular+GNSS in one element
Limitations	Needs the largest keep-out area of any antenna in the design; performance is sensitive to how it is folded/routed in the final case
Estimated Price	\$1.50–\$2.50 (1k units)
Datasheet Reference	taoglas.com — FXUB66 Datasheet

SOS Emergency Button

Field	Detail
Manufacturer	Alps Alpine
Part Number	Low-profile waterproof SMD tactile switch (SKRG-series class)
Category	User Input
Function	Physical, unmistakable manual trigger for an emergency alert — independent of any AI decision.

Field	Detail
Why Selected	A dedicated physical button provides a fail-safe manual override that works even if sensors or AI misclassify a situation; the waterproof SMD variant preserves the device IP rating without a mechanical boot.
Package	Low-profile waterproof SMD tact switch
Dimensions	≈4.5 × 4.5 × 1.5 mm (typical low-profile package)
Voltage	Logic-level (3.0–3.3V) switched to GPIO
Power Consumption	Negligible — only pull current when pressed
Communication Interface	GPIO with internal/external pull-up, debounced in firmware
Typical Applications	Wearables, remotes, medical alert pendants
Advantages	IP-rated sealing, tactile confirmation click, long mechanical life
Limitations	Purely mechanical — must be firmware-debounced to avoid false triggers
Estimated Price	\$0.30–\$0.60 (1k units)
Datasheet Reference	alps.alpine.com — SKRG-series datasheet

Magnetic Pogo Pin Connector

Field	Detail
Manufacturer	Connector vendor (custom/standard 2-pin magnetic pogo)
Part Number	2-Pin Magnetic Pogo Connector Set
Category	Charging Interface
Function	Provides a tool-free, waterproof-friendly charging connection between an external charging dock/cable and the PMIC input.
Why Selected	Magnetic pogo pins allow charging without any exposed port (no USB-C cutout to seal), which is the standard approach for IP-rated wearables and keeps the enclosure fully sealed.
Package	2-pin spring-loaded pogo pin pair + matching magnetic pad
Dimensions	≈4–6 mm pin diameter, contact pitch application-specific
Voltage	5V input (typical charge adapter output)
Power Consumption	Up to ≈500mA–1A charge current capability (connector-rated)
Communication Interface	Direct 2-wire (VBUS/GND) to nPM1300 charge input
Typical Applications	Smartwatches, wearables, medical wearables
Advantages	Fully sealed charging (no port), magnetic self-alignment, durable
Limitations	Exposed metal contacts need ESD/reverse-polarity protection on the PCB side
Estimated Price	\$0.80–\$1.50 per set (1k units)
Datasheet Reference	Connector vendor mechanical drawing + plating spec (typ. gold-plated contacts)

5. Component Deep Explanation

This section explains each component from first principles — what it is, the problem it solves, how it works internally, and what would happen if it were removed. It is written for an engineer moving from software/cybersecurity into embedded hardware, so each explanation assumes no prior hardware background.

nRF5340 SoC — Deep Explanation

What is it?

The nRF5340 is a "System-on-Chip" (SoC): a single piece of silicon containing not one but two independent ARM Cortex-M33 processors, radio hardware, memory, and dozens of peripheral interfaces (I2C, SPI, UART, etc.) all in one package. Think of it as two small computers sharing one chip: an "Application" core for your logic/AI, and a "Network" core dedicated purely to running the Bluetooth radio.

What problem does it solve, and why is it necessary?

A wearable needs to do two things simultaneously and reliably: (1) run continuous sensor-fusion and AI inference, and (2) maintain a rock-solid Bluetooth Low Energy connection with strict timing requirements. On a single-core MCU, a heavy AI computation can stall the radio timing and cause dropped BLE connections. The nRF5340 solves this by physically separating the two jobs onto two cores that run independently.

How does it work internally?

The Application core (Cortex-M33 @128MHz) executes your main firmware — sensor drivers, the decision engine, and AI inference. It communicates internally with the Network core (Cortex-M33 @64MHz) over a fast internal bus; the Network core runs Nordic's pre-certified BLE protocol stack and talks directly to the on-chip 2.4GHz radio transceiver, which up-converts/down-converts baseband signals to/from the RF antenna pin.

How does it communicate?

Externally, the Application core exposes standard buses (I2C, SPI, UART, PDM, GPIO) used to talk to every sensor in the design. The RF radio pin connects through a matching network to the Johanson ceramic BLE antenna.

Where is it placed inside the wearable, and why?

Centrally located on the PCB, roughly equidistant from all sensors to keep bus trace lengths short, but positioned so its RF pin has a short, clean path to the BLE antenna at the board edge. Short traces reduce signal noise and parasitic inductance on both the digital buses and — critically — the RF feed line, where trace length directly affects impedance matching and radiated efficiency.

What would happen if it were removed?

Without the nRF5340, there is no system at all — no sensor reading, no AI decision-making, no Bluetooth link, and no way to hand data to the cellular modem. It is the mandatory central nervous system of the device.

Required External Components

The following passive/support components are mandatory around this part, per the manufacturer reference design:

Component	Qty	Recommended Value	Package	Function
32 MHz crystal (HFXO)	1	±10-20ppm, CL=9-12pF (per datasheet)	3225 (3.2×2.5mm)	High-accuracy clock for the radio and CPU core timing
32.768 kHz crystal (LFXO)	1	±20ppm, CL=9-12.5pF	3215 or 1610	Low-power sleep-mode / RTC timekeeping clock
Crystal load capacitors	4	9-12pF (matched to crystal CL)	0402	Set correct oscillation frequency for both crystals
Decoupling capacitors (VDD)	6-8	100nF	0402	Local high-frequency noise filtering at each power pin
Bulk decoupling capacitor	2	4.7μF–10μF	0603	Smooths transient current draw during radio TX bursts
DC/DC buck inductor	1	10μH (if internal REG1 DC/DC enabled)	2016/0806	Required only if using nRF5340 internal buck regulator mode for efficiency
SWD programming header	1	2×5 1.27mm pitch (Tag-Connect or pin header)	THT	Firmware flashing and JTAG/SWD debug access

nRF9151 SiP — Deep Explanation

What is it?

The nRF9151 is a "SiP" (System-in-Package): unlike the nRF5340's single silicon die, it packages a full LTE-M/NB-IoT cellular modem chip, an application-grade ARM Cortex-M33, RF front-end components, and a power management unit together as one solderable part — essentially a complete miniature cell phone modem.

What problem does it solve, and why is it necessary?

Bluetooth only works within roughly 10-30 meters of a paired phone. If the wearer falls without their phone nearby (asleep, phone in another room, phone dead), a BLE-only device cannot call for help. The nRF9151 gives the wearable its own independent connection to the cellular network — the same technology (LTE) your phone uses to make calls without Wi-Fi.

How does it work internally?

Internally the SiP contains a baseband modem (handles the complex mathematics of encoding/decoding cellular radio signals), an RF transceiver, a power amplifier for transmission, and its own small application processor that runs the cellular protocol stack and exposes a simple AT-command interface — the same command language dial-up modems used decades ago, still the industry standard for cellular modules.

How does it communicate?

The nRF5340 talks to the nRF9151 over a simple UART serial link, sending text commands like "connect to network" or "send this GPS+alert payload" and receiving responses. The nRF9151 has its own separate RF pin routed to the Taoglas flexible antenna.

Where is it placed inside the wearable, and why?

Placed toward the opposite edge of the PCB from the BLE antenna/radio, with its own RF keep-out zone, to minimize any electromagnetic interference between the two independent radios operating in different frequency bands. Two active radios in a very small product can desensitize each other if placed too close together; separating them and giving each its own ground/keep-out area is standard RF coexistence practice.

What would happen if it were removed?

Without the nRF9151, the device becomes entirely dependent on a paired phone for any emergency alert — a single point of failure that defeats the product's core safety promise.

Required External Components

The following passive/support components are mandatory around this part, per the manufacturer reference design:

Component	Qty	Recommended Value	Package	Function
Decoupling capacitors	8-10	100nF + 1 μ F mix	0402/0603	Filters supply noise, especially during high-current TX bursts
Bulk/reservoir capacitor	1-2	47 μ F–100 μ F, low-ESR	1206 or tantalum	Buffers the ~500mA current spikes during LTE transmit so the battery rail does not sag
SIM card holder or eSIM (MFF2)	1	Nano-SIM socket or embedded eSIM chip	THT/SMD	Cellular network authentication (carrier/eSIM profile)
RF matching network (cellular)	~4 components	L/C per Nordic reference design	0402	Impedance-matches nRF9151 RF pin to 50 Ω antenna feed
RF matching network (GNSS)	~3 components	L/C per Nordic reference design	0402	Matches the separate GNSS receive path to the antenna
ESD protection diode (antenna line)	1	Low-capacitance ESD array, <0.5pF	0402	Protects RF front-end from static discharge via the external antenna

BMI270 IMU — Deep Explanation

What is it?

The BMI270 is an Inertial Measurement Unit (IMU): a chip containing a tiny mechanical accelerometer (measures linear acceleration/gravity in 3 axes) and a gyroscope (measures rotational velocity in 3 axes) fabricated using MEMS (Micro-Electro-Mechanical Systems) technology — literally microscopic mechanical structures etched into silicon.

What problem does it solve, and why is it necessary?

Fall detection and activity classification both require knowing how the wrist is moving in 3D space, at high enough sample rates to distinguish a genuine fall (sharp free-fall followed by an impact spike) from normal activity (walking, typing, exercising).

How does it work internally?

Inside the MEMS structure, tiny suspended proof masses shift position under acceleration or rotation, changing a measurable capacitance; on-chip analog and digital circuitry converts that capacitance change into a precise digital value for each axis. The BMI270 additionally has a small internal state machine capable of running basic motion-classification algorithms (step counting, tap detection) without waking the main MCU.

How does it communicate?

Connects over I2C to the nRF5340's sensor bus, and raises a hardware interrupt (INT1) directly to a GPIO pin the instant a "significant motion" or free-fall condition is detected — the MCU does not need to continuously poll it.

Where is it placed inside the wearable, and why?

As close as practical to the wearable's geometric/rotational center, so the measured motion best represents the whole device's movement rather than being dominated by local vibration at one edge. Placing an IMU off-center introduces a "lever arm" effect where rotation produces extra apparent linear acceleration, subtly corrupting fall-detection accuracy.

What would happen if it were removed?

Without the BMI270, the device loses all fall detection and activity classification — arguably the single most important safety feature of the product.

Required External Components

The following passive/support components are mandatory around this part, per the manufacturer reference design:

Component	Qty	Recommended Value	Package	Function
Decoupling capacitor (VDD)	1	100nF	0201/0402	Digital supply noise filtering
Decoupling capacitor (VDDIO)	1	100nF	0201/0402	I/O supply noise filtering
I2C pull-up resistors (SDA/SCL)	2	2.2k Ω –4.7k Ω	0402	Required on every I2C bus — open-drain lines need pull-ups to VDDIO

MAX86176 Health AFE — Deep Explanation

What is it?

The MAX86176 is an Analog Front-End (AFE): a chip that bridges the physical, analog world (light reflected off skin, tiny electrical signals from the heart) and the digital world the MCU understands. It combines LED drivers, ultra-sensitive light-sensing amplifiers, and an ECG amplifier with a precision analog-to-digital converter (ADC).

What problem does it solve, and why is it necessary?

Measuring heart rate optically (PPG — photoplethysmography) requires shining light into the skin and measuring tiny variations in reflected light caused by blood volume changes with each heartbeat — a signal thousands of times smaller than the ambient noise around it. This requires specialized, low-noise analog circuitry that a general-purpose MCU does not have.

How does it work internally?

The chip pulses one or more LEDs (typically green and infrared) at a controlled current, then a photodiode-facing amplifier measures the tiny fraction of light reflected back from blood vessels under the skin. A high-resolution ADC digitizes this waveform. A separate ECG front-end amplifies the millivolt-level electrical signal generated by the heart muscle, detected through skin-contact electrodes.

How does it communicate?

Uses SPI (a fast 4-wire bus) to stream continuous waveform samples to the nRF5340, since PPG/ECG data rates are higher than a simple I2C sensor like the temperature chip. An interrupt line signals "new sample ready" or "finger/skin contact lost."

Where is it placed inside the wearable, and why?

Mounted directly beneath an optical window on the underside of the wearable, in continuous contact with the wearer's skin — this is non-negotiable for an optical sensor. PPG signal quality degrades enormously with any air gap or light leakage between the LED/photodiode and the skin — the entire mechanical design of the case bottom exists to seal this optical path.

What would happen if it were removed?

Without the MAX86176, the device loses heart-rate and cardiac-anomaly detection — one of the two biometric "abnormal pattern" triggers described in the project brief.

Required External Components

The following passive/support components are mandatory around this part, per the manufacturer reference design:

Component	Qty	Recommended Value	Package	Function
LED driver decoupling capacitor	1-2	1 μ F–2.2 μ F, low-ESR	0402/0603	Supplies instantaneous current for LED drive pulses without sagging the rail
Analog supply decoupling	2	100nF + 1 μ F	0402	Filters analog front-end supply for low-noise PPG signal
Green/Red/IR PPG LEDs	2-3	Per optical stack-up (commonly green + IR, or green/red/IR)	0603 LED package	Light sources for photoplethysmography
Photodiode	1	High-sensitivity, matched to LED wavelength	SMD optical	Detects reflected light to derive the PPG waveform
ECG lead input protection/filtering	2-3	Series resistor + ESD diode per lead	0402	Protects ECG front-end and reduces high-frequency noise
Optical light-block gasket	1	Custom molded, opaque	Mechanical	Prevents ambient light leakage from corrupting the PPG signal

MAX30208 Temp Sensor — Deep Explanation

What is it?

The MAX30208 is a fully digital temperature sensor: unlike a simple analog thermistor, it contains its own tiny temperature-sensing element, ADC, and digital logic, and directly outputs a calibrated temperature reading in degrees Celsius over I2C — no external calibration math required.

What problem does it solve, and why is it necessary?

Detecting fever, hypothermia, or heat-stroke risk requires continuous, clinically-accurate skin temperature monitoring, and the system needs to be alerted immediately if temperature crosses a dangerous threshold — ideally without waking the main CPU to constantly check.

How does it work internally?

A bandgap-reference temperature-sensing circuit produces a voltage proportional to absolute temperature; an internal ADC converts this to a digital code, which is scaled to a 0.005°C-resolution reading. A programmable comparator continuously checks this value against user-set high/low thresholds in hardware.

How does it communicate?

I2C bus, shared with other low-speed sensors. Its ALERT pin is wired directly to a GPIO and fires a hardware interrupt the instant temperature crosses a configured threshold — an efficient, CPU-free way to catch an emergency temperature event.

Where is it placed inside the wearable, and why?

Mounted on the skin-facing side of the PCB, close to the MAX86176 optical sensor, so both share the same skin-contact window/thermal path. Any air gap between the sensor and skin, or proximity to a heat-generating component (like the LTE power amplifier), would corrupt the temperature reading with a systemic offset.

What would happen if it were removed?

Without the MAX30208, the device cannot detect abnormal-temperature emergencies (e.g., heat stroke, hypothermia, high fever) — one of the explicitly required AI-monitored conditions.

Required External Components

The following passive/support components are mandatory around this part, per the manufacturer reference design:

Component	Qty	Recommended Value	Package	Function
Decoupling capacitor	1	100nF	0201/0402	Local supply noise filtering
I2C pull-up resistors	2 (shared bus)	2.2kΩ–4.7kΩ	0402	Shared with other I2C devices on the same bus if same rail/voltage domain

IM69D130 MEMS Microphone — Deep Explanation

What is it?

The IM69D130 is a MEMS (Micro-Electro-Mechanical Systems) digital microphone: a microscopic mechanical diaphragm etched in silicon vibrates in response to sound pressure, and on-chip electronics convert that vibration

directly into a digital bitstream — no separate microphone preamp or analog-to-digital converter chip is needed elsewhere on the board.

What problem does it solve, and why is it necessary?

Some emergencies (a scream, shouting for help, the sound of a crash) are audible before — or instead of — being visible in motion or heart-rate data. The system needs an acoustic sensing channel, and it needs one with enough dynamic range to hear a quiet distress call without clipping during a sudden loud event.

How does it work internally?

The mechanical MEMS diaphragm forms one plate of a capacitor; sound pressure moves the diaphragm, changing the capacitance. An on-chip ASIC converts this changing capacitance into a high-frequency 1-bit digital stream using a technique called PDM (Pulse Density Modulation) — the density of 1s vs 0s over time represents the audio waveform.

How does it communicate?

PDM bus: just two wires, a clock line (driven by the nRF5340) and a data line (driven by the microphone). The nRF5340's hardware PDM peripheral converts this bitstream into normal PCM audio samples in the background with almost no CPU load.

Where is it placed inside the wearable, and why?

Positioned near the case perimeter, aligned with a small acoustic port/mesh in the housing, on the side of the wearable most likely to face away from the wrist (i.e., outward, not pressed against skin). A microphone sealed too far from an acoustic opening, or muffled against skin/fabric, loses high-frequency sensitivity and overall sound-pressure accuracy.

What would happen if it were removed?

Without the microphone, the device loses its acoustic emergency-detection channel — it would miss distress situations that are audible before they show up in motion or vital-sign data.

Required External Components

The following passive/support components are mandatory around this part, per the manufacturer reference design:

Component	Qty	Recommended Value	Package	Function
Decoupling capacitor	1	100nF	0201/0402	Local supply filtering close to the mic pin
Acoustic mesh / gasket	1	IP-rated acoustic mesh	Mechanical	Protects the acoustic port from water/dust while preserving sound path

nPM1300 PMIC — Deep Explanation

What is it?

The nPM1300 is a Power Management IC (PMIC): a single chip that replaces what would otherwise be five or six separate chips — a battery charger, multiple voltage regulators, and a fuel gauge — all coordinated together and controllable by the main MCU over I2C.

What problem does it solve, and why is it necessary?

A wearable has many components needing different, precisely regulated voltages (1.8V for the MCU core, 3.0V for the modem, etc.), all powered from a single 3.7V battery that itself needs safe, monitored charging. Doing this with discrete regulators would take far more PCB space than a slim wristband allows.

How does it work internally?

The charger block manages the constant-current/constant-voltage (CC/CV) charging algorithm required to safely charge lithium-polymer cells, monitoring voltage, current, and (via an external thermistor) temperature to prevent unsafe conditions. Two on-chip buck (step-down) converters efficiently generate lower voltage rails from the battery, and two LDOs (linear regulators) provide clean, low-noise power for sensitive analog circuits. A coulomb-counting fuel gauge tracks charge in/out of the battery to estimate remaining capacity accurately.

How does it communicate?

Controlled and monitored by the nRF5340 over I2C — the MCU can query battery percentage, enable/disable individual rails (to save power when a sensor isn't needed), and receive interrupts for charge-complete or fault conditions.

Where is it placed inside the wearable, and why?

Located close to both the battery connector and the bulk of the digital ICs it powers, minimizing the length of high-current power traces. Long power traces increase resistive (IR) voltage drop and can radiate switching noise from the buck converters into sensitive nearby analog circuits (like the MAX86176) if not placed and routed carefully.

What would happen if it were removed?

Without the PMIC, there is no safe way to charge the battery or generate the multiple voltage rails the design needs — the system simply cannot power on.

Required External Components

The following passive/support components are mandatory around this part, per the manufacturer reference design:

Component	Qty	Recommended Value	Package	Function
Buck converter inductors	2	1.0μH–2.2μH, shielded	2016/2520	Energy-storage element for each buck regulator output
Buck output capacitors	2	10μF–22μF	0603/0805	Output ripple filtering for each buck rail
LDO output capacitors	2	1μF	0402	Stability/filtering capacitor required by each LDO
VBAT input bulk capacitor	1	10μF	0603	Buffers battery-side transients (esp. during LTE TX bursts)

Component	Qty	Recommended Value	Package	Function
Charge-current sense / thermistor	1	NTC 10kΩ (battery pack integrated or external)	THT/SMD	Battery temperature monitoring during charge for safety (JEITA profile)
I2C pull-up resistors	2	2.2kΩ–4.7kΩ	0402	Required for the PMIC control/telemetry I2C bus

Li-Po Battery Pack — Deep Explanation

What is it?

A lithium-polymer (Li-Po) battery stores energy chemically in a flexible pouch cell, rather than the rigid metal can of a cylindrical lithium-ion cell — allowing it to be shaped to fit the curved, slim geometry of a wristband.

What problem does it solve, and why is it necessary?

Every active component in this document ultimately draws its power from somewhere; the wearable needs a rechargeable energy source dense and thin enough to disappear inside a slim wrist-worn enclosure while still lasting through a full day (or more) including periodic LTE transmissions.

How does it work internally?

Inside the pouch, alternating layers of lithium-cobalt-oxide (or similar) cathode, graphite anode, separator, and electrolyte are stacked or wound and sealed in a foil pouch. Lithium ions move between the electrodes during charge/discharge, releasing/storing electrical energy.

How does it communicate?

Purely a power-delivery element — no digital communication itself. The PMIC monitors it indirectly (voltage, current, and, via the pack's integrated thermistor, temperature).

Where is it placed inside the wearable, and why?

Occupies the single largest area on the PCB stack (see Section 6), typically the geometric center-to-wide-face of the wristband puck, since energy capacity scales directly with physical volume. Placing the battery away from the antennas avoids detuning them (batteries contain conductive foil that disrupts RF fields), and keeping it away from the LTE power amplifier avoids extra thermal stress during charge.

What would happen if it were removed?

Without a battery, the device has no energy source and cannot function at all as a portable wearable.

Required External Components

The following passive/support components are mandatory around this part, per the manufacturer reference design:

Component	Qty	Recommended Value	Package	Function
Protection Circuit Module (PCM)	1 (often built)	Over-voltage / over-current / short-circuit cutoff	Integrated FPC	Mandatory safety circuit — prevents overcharge, over-

Component	Qty	Recommended Value	Package	Function
	into the cell pack)			discharge, and short-circuit damage/fire risk
NTC thermistor (pack-integrated)	1	10kΩ B25/50	Integrated in pack	Temperature feedback to the PMIC charge controller
Battery connector or solder pads	1	2-pin JST-SH 1.0mm or pads	SMD/THT	Mechanical/electrical connection between the cell and the main PCB

LRA Haptic Actuator — Deep Explanation

What is it?

A Linear Resonant Actuator (LRA) is a small motor-like component containing a magnetic mass suspended on a spring, driven back and forth by a coil at a specific resonant frequency — unlike the spinning off-center weight of an older "vibration motor," an LRA moves in a straight line, producing a sharper, more precisely controllable vibration.

What problem does it solve, and why is it necessary?

The wearable has no screen, so haptic (vibration) feedback is one of its only ways to communicate directly with the wearer — confirming an alert was sent, warning of an abnormal reading, or acknowledging an SOS press.

How does it work internally?

A voice-coil-like electromagnet inside the actuator alternately attracts and repels a spring-mounted magnetic mass; driven at exactly the system's mechanical resonant frequency, small drive currents produce surprisingly strong, crisp vibration — but only within a narrow frequency band, unlike a simple DC motor.

How does it communicate?

Not driven directly by the MCU's GPIO (a raw digital pin cannot supply the correct resonant AC waveform). Instead, a dedicated haptic driver IC (see Required External Components below), controlled over I2C/PWM by the nRF5340, synthesizes the correct waveform and can even auto-detect the actuator's exact resonant frequency.

Where is it placed inside the wearable, and why?

Mounted against the case wall on the side that best transmits vibration to the wrist, mechanically decoupled from the main PCB where possible to avoid vibrating sensitive components like the IMU. A poorly coupled or loosely mounted actuator wastes vibration energy into the case instead of the wearer's skin, and can inject false motion signals into the nearby BMI270 if placed too close.

What would happen if it were removed?

Without the haptic actuator, the device has no way to give the wearer any feedback at all — it becomes a "silent" device that provides no local confirmation of alerts.

Required External Components

The following passive/support components are mandatory around this part, per the manufacturer reference design:

Component	Qty	Recommended Value	Package	Function
Haptic driver IC (e.g., TI DRV2605L-class)	1	I2C-controlled haptic driver with auto-resonance tracking	QFN/DSBGA	Generates the correct resonant AC waveform for the LRA — a bare GPIO cannot drive an LRA correctly
Driver output decoupling capacitor	1	1 μ F	0402	Local supply filtering for the haptic driver
Series/damping resistor (if applicable)	1	Per driver reference design	0402	Waveform shaping / current limiting on the drive line

BLE Ceramic Chip Antenna — Deep Explanation

What is it?

A ceramic chip antenna is a small block of ceramic dielectric material with a metallized conductive pattern on its surface, engineered so that its physical dimensions resonate at 2.4GHz — the Bluetooth frequency — turning an electrical signal into a radiated radio wave (and vice-versa on receive).

What problem does it solve, and why is it necessary?

The nRF5340's BLE radio can generate a perfect 2.4GHz electrical signal internally, but that signal needs a physical structure to actually radiate into the air as a usable radio wave that a phone can receive tens of meters away.

How does it work internally?

The ceramic's high dielectric constant allows the antenna structure to be physically much smaller than a bare copper antenna would need to be at the same frequency (a "wavelength-shrinking" effect), which is why it can be as small as a rice grain while still being reasonably efficient.

How does it communicate?

Not a communication interface in the digital sense — it is fed by a 50-ohm RF trace from the nRF5340's RF pin, through a matching network that maximizes power transfer between the chip and the antenna.

Where is it placed inside the wearable, and why?

At the PCB edge, farthest from the battery and any large ground-plane copper pour, inside a defined RF keep-out zone (see Section 11). Any nearby metal or ground copper detunes the antenna's resonant frequency and reduces its radiated efficiency — keep-out zones exist specifically to protect this.

What would happen if it were removed?

Without this antenna, the BLE radio has no way to radiate its signal — the phone app, local sync, and short-range alerts would all stop working, though the LTE fallback would still function.

Required External Components

The following passive/support components are mandatory around this part, per the manufacturer reference design:

Component	Qty	Recommended Value	Package	Function
RF matching network (Pi: L-C-L or C-L-C)	3	Tuned per Johanson application note (typ. 0402 values, e.g. 1.5nH-2.2pF-1.5nH — to be tuned on real board)	0402	Matches the antenna impedance to the 50Ω radio output for maximum power transfer
RF feed trace	1	50Ω controlled-impedance microstrip/CPW	PCB trace	Carries the RF signal from IC to antenna with minimal loss/reflection

LTE/GNSS Flexible Antenna — Deep Explanation

What is it?

The Taoglas FXUB66 is a flexible printed antenna on a thin polymer film with an adhesive backing, engineered to cover the wide range of frequency bands used globally by LTE-M/NB-IoT networks, plus the separate GNSS receive band, all in one physical element.

What problem does it solve, and why is it necessary?

Cellular and GNSS frequencies are much lower than Bluetooth's 2.4GHz, which normally requires a physically larger antenna — a serious challenge in a slim wristband. A flexible design lets the antenna be folded or curved along the inside of the case wall to fit the available space.

How does it work internally?

The antenna's conductive trace pattern is printed on a thin flexible substrate; its geometry is designed with multiple resonant sections so it efficiently covers several widely separated frequency bands (cellular low/mid bands plus the ~1.5-1.6GHz GNSS band) simultaneously.

How does it communicate?

Passive RF element, connected via a coax pigtail or direct RF trace to the nRF9151's antenna pin, through a matching/diplexer network that separates the cellular and GNSS signal paths.

Where is it placed inside the wearable, and why?

Routed along the inner wall of the wearable case, on the opposite side of the PCB from the BLE antenna, inside its own (larger) RF keep-out zone. This antenna needs the most electrically "clean" space in the whole product — it is the literal lifeline for an emergency alert when Bluetooth is unavailable, so its performance is treated as a first-class design priority, not an afterthought.

What would happen if it were removed?

Without this antenna, the nRF9151 modem cannot transmit or receive at all, eliminating the device's entire phone-independent emergency-communication capability and its GNSS location feature.

Required External Components

The following passive/support components are mandatory around this part, per the manufacturer reference design:

Component	Qty	Recommended Value	Package	Function
U.FL / MHF coax connector or pigtail	1	Micro coaxial, 50Ω	SMD	Connects the flexible antenna to the PCB RF trace with minimal loss
RF matching/diplexer network	2-4	Per Taoglas application note	0402	Separates and matches the cellular and GNSS frequency paths
ESD protection diode	1	Low-capacitance, <0.5pF	0402	Protects the modem RF input from static discharge entering via the antenna

SOS Emergency Button — Deep Explanation

What is it?

A tactile switch is a simple mechanical component: pressing it physically closes an electrical contact, connecting two pins together. The "waterproof SMD" variant adds an internal rubber/silicone seal so the button can be pressed from outside a sealed case without breaking water resistance.

What problem does it solve, and why is it necessary?

AI systems can misclassify situations, and sensors can occasionally fail silently. A safety-critical wearable needs one guaranteed, human-operated way to signal "I need help right now" that does not depend on any software decision being correct.

How does it work internally?

Beneath the button cap, a metal dome or rubber contact bridges two PCB pads when pressed, completing a circuit that pulls a GPIO pin from its resting logic level to the opposite level. The waterproof sealing is purely mechanical (an internal gasket), not electrical.

How does it communicate?

A single GPIO pin, held at a stable logic level by a pull-up resistor, with the switch pulling it to the opposite level when pressed. Firmware debounces the signal (ignores rapid electrical "chatter" from the mechanical contact) before treating it as a genuine press.

Where is it placed inside the wearable, and why?

On the side or top edge of the case where it is reachable but not so exposed that it could be triggered accidentally (e.g., not directly under where a sleeve cuff would press). Placement balances two competing needs: it must be easy to find and press quickly under stress, but resistant to accidental activation during normal daily wear.

What would happen if it were removed?

Without the SOS button, the wearer loses their one guaranteed manual override — every emergency trigger would then depend entirely on AI/sensor correctness, which is an unacceptable single point of failure for a safety product.

Required External Components

The following passive/support components are mandatory around this part, per the manufacturer reference design:

Component	Qty	Recommended Value	Package	Function
Pull-up resistor	1	10kΩ	0402	Defines a stable logic-high level when the button is not pressed
Debounce/EMI filter capacitor	1	100nF (or 10-47nF for fast filtering)	0402	Filters mechanical contact bounce and ESD/EMI spikes on the GPIO line
ESD protection diode (if exposed to case exterior)	1	Low-capacitance TVS	0402	Protects the GPIO pin from static discharge through the exposed button

Magnetic Pogo Pin Connector — Deep Explanation

What is it?

A magnetic pogo pin connector is a pair of small spring-loaded metal pins (they compress slightly like a tiny pen spring when pressed) paired with a magnet that pulls a matching charging dock into perfect alignment automatically.

What problem does it solve, and why is it necessary?

A wearable with a traditional USB-C port needs a port cover or precise gasket to stay water-resistant, adding cost, complexity, and a failure point. A fully sealed, portless design needs some other way to get charging current in.

How does it work internally?

Each pogo pin is a small barrel containing a spring and a plunger tip; when the magnetic charging cable's matching flat contacts touch the plungers, they compress slightly, guaranteeing reliable electrical contact even with minor misalignment, while the embedded magnets pull the cable into the correct position automatically.

How does it communicate?

A simple 2-wire (power + ground) DC connection — not a digital interface. Charging current flows directly from the dock, through the pins, into the PMIC's charge input.

Where is it placed inside the wearable, and why?

On the flat underside or edge of the case, positioned so it does not overlap with the skin-contact sensor windows and remains accessible while worn on the wrist. The exposed metal contacts are also the device's only point of contact with an external, less-trusted electrical source, so they are placed with a clear, protected trace path back to the PMIC's protection circuitry.

What would happen if it were removed?

Without a charging connector, the battery could never be recharged once depleted, making the device effectively disposable.

Required External Components

The following passive/support components are mandatory around this part, per the manufacturer reference design:

Component	Qty	Recommended Value	Package	Function
Reverse-polarity protection (P-FET or diode)	1	Per PMIC input current rating	SOT-23	Prevents damage if the charge cable is connected with reversed polarity
TVS/ESD diode on VBUS line	1	5.5V standoff, low clamping voltage	0402/SOD-523	Protects the charge input from ESD entering via the exposed pogo contacts
Input filter capacitor	1	1 μ F–10 μ F	0603	Filters charger-side noise before it reaches the PMIC

6. PCB Placement

The floor plan below shows the approximate relative placement and sizing of every major component on the main PCB, viewed from the top. Placement decisions are driven by three competing priorities: RF performance (antenna keep-out), sensor accuracy (skin contact for optical/temperature sensors), and mechanical/thermal separation (battery away from RF, haptic motor away from the IMU).

PCB Component Placement — Top View (Wristband Puck, ~42mm x 30mm)



Figure 6.1 — Top-down PCB placement floor plan.

6.1 Placement Rationale

Component	Zone	Reason
nRF5340	Board center	Short, balanced trace lengths to all sensors and to both radios
nRF9151	Opposite edge from BLE radio	RF isolation between the two independent radios
BMI270	Near geometric/rotational center	Most representative motion reading, minimal lever-arm error
MAX86176 + MAX30208	Underside, skin-contact window	Optical/thermal sensors must directly contact skin
IM69D130 Microphone	Case perimeter, acoustic port	Needs an open path to ambient sound, away from skin seal
nPM1300	Near battery connector	Minimizes high-current trace length/resistive loss
Battery	Largest open board/case area	Capacity scales with volume; kept away from antennas

Component	Zone	Reason
LRA Haptic	Case wall, away from IMU	Direct skin coupling; avoids injecting vibration noise into BMI270
BLE Antenna	PCB edge, dedicated keep-out	Requires clear space free of ground copper/metal
LTE/GNSS Antenna	Opposite PCB edge, largest keep-out	Widest frequency range, most sensitive to detuning
SOS Button	Top/side case edge	Reachable, but resistant to accidental press
Pogo Charging Pins	Flat underside/edge, away from sensor windows	External access point, isolated from sensitive analog zones

7. Internal Exploded View

This illustration shows every internal layer of the wearable stacked in assembly order, from the outward-facing lens down to the wristband strap. Understanding this stack-up matters even before schematic capture begins, because it constrains component height (Z-axis) budgets on the PCB.

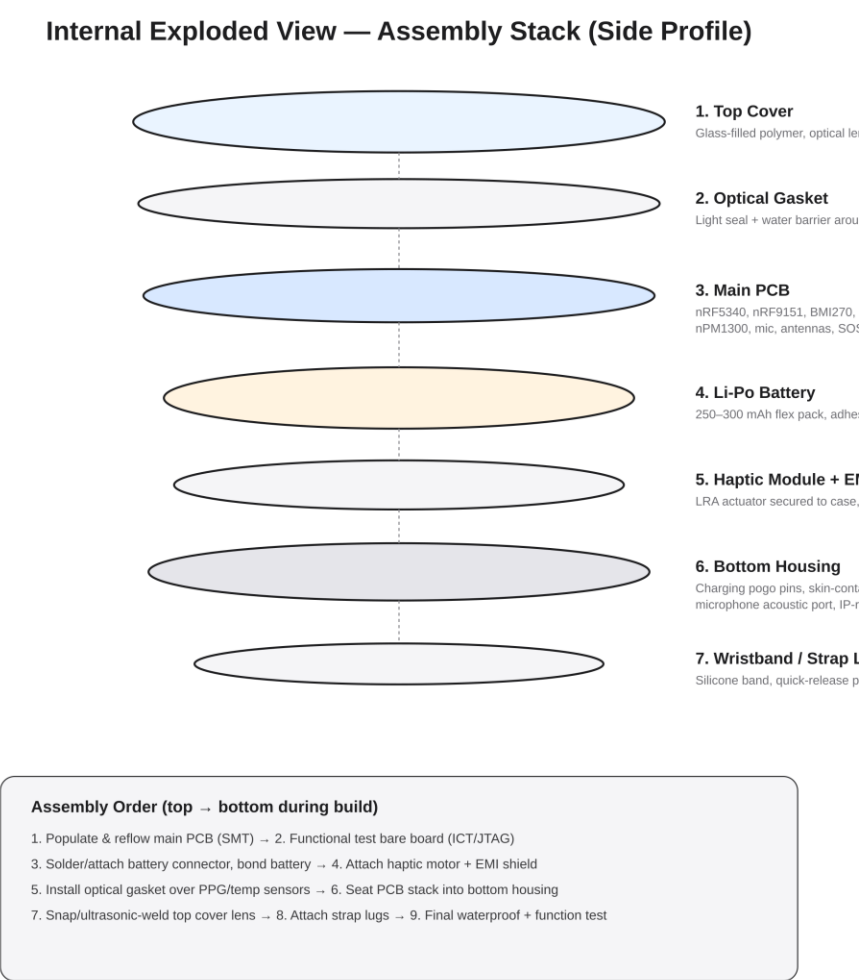


Figure 7.1 — Exploded assembly stack, top cover to strap.

8. Signal Flow

The diagram below traces a single sensor reading from the moment it is sampled through to the moment a caregiver sees it in the mobile app — including the branch point where the system decides between a quiet background log, a local haptic alert, or a full emergency transmission.

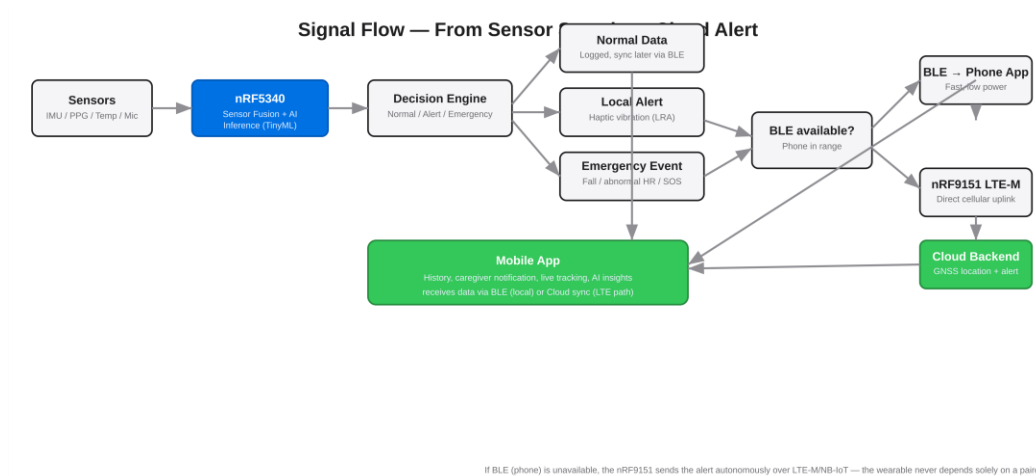


Figure 8.1 — End-to-end signal flow: sensors → decision → BLE/LTE → cloud → app.

8.1 Decision Logic Summary

- Normal data: logged locally, opportunistically synced to the phone app over BLE — no urgency, no wake of the LTE modem (saves power).
- Local alert (e.g., mildly elevated heart rate): triggers a haptic vibration immediately; also queued for BLE sync.
- Emergency event (fall, abnormal HR/temperature, SOS press, distress sound): the nRF5340 immediately attempts BLE delivery to a paired phone; if no phone responds within a short timeout, it powers up the nRF9151 and sends the alert plus GNSS location over LTE-M/NB-IoT directly to the cloud backend.

9. Power Architecture

Every component in this design ultimately draws current from a single 3.7V Li-Po cell. The nPM1300 PMIC is responsible for turning that single voltage into every rail the system needs, safely charging the battery, and reporting power status back to the MCU.

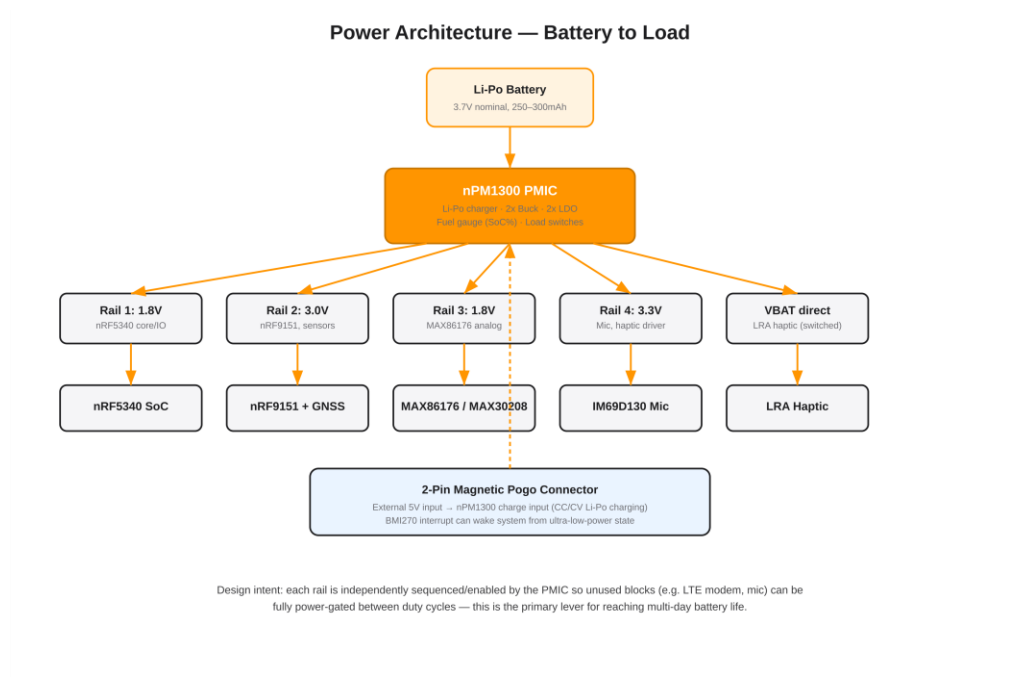


Figure 9.1 — Power distribution tree: battery → PMIC → rails → loads.

9.1 Rail Summary

Rail	Approx. Voltage	Feeds	Type
Rail 1	1.8V	nRF5340 core/IO logic	Buck (switching, efficient)
Rail 2	3.0V	nRF9151 modem, digital sensors	Buck (switching, efficient)
Rail 3	1.8V	MAX86176 analog front-end	LDO (low-noise, linear)
Rail 4	3.3V	Microphone, haptic driver	LDO (low-noise, linear)
VBAT direct	3.0–4.2V (battery voltage)	Haptic actuator (switched)	Direct, load-switched

Note: Analog-sensitive rails (MAX86176) are deliberately fed by low-noise LDOs rather than switching bucks, since buck-converter switching noise can corrupt a microvolt-level PPG signal.

10. Communication Interfaces

This section explains, in plain terms, every communication protocol used in the design — what it physically is, and why it was chosen for each connection.

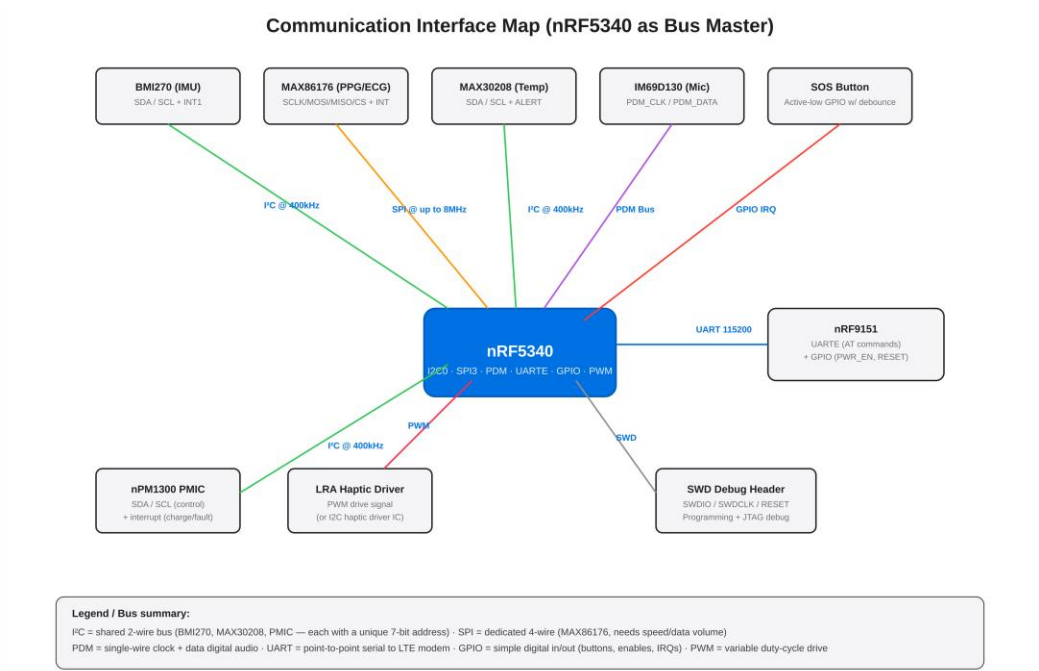


Figure 10.1 — Full communication interface map, nRF5340 as bus master.

Protocol	What it physically is	Used for
I ² C	Two-wire (SDA/SCL) shared bus; multiple devices share the same two wires, each with a unique address.	BMI270, MAX30208, nPM1300 — low/medium-speed sensors and control interfaces where sharing wires saves pins and PCB space.
SPI	Four-wire (SCLK/MOSI/MISO/CS) point-to-point bus; faster than I2C, one dedicated chip-select per device.	MAX86176 — needs higher data throughput for continuous PPG/ECG waveform streaming.
UART	Simple two-wire (TX/RX) point-to-point serial link, no shared clock.	nRF5340 ↔ nRF9151 — the modem exposes a standard AT-command serial interface, the industry-standard way to control cellular modems.
GPIO	General-Purpose Input/Output — a simple digital pin, high or low.	SOS button, sensor interrupt lines, enable/reset lines for the modem.
PWM	Pulse-Width Modulation — a digital signal whose "on" time is varied to represent an analog-like effect.	Drive signal input to the haptic driver IC (which converts it into the LRA's resonant AC waveform).
PDM	Pulse-Density Modulation — a 1-bit high-frequency digital audio stream (clock + data).	IM69D130 digital microphone — avoids needing a separate analog mic preamp and ADC.
BLE	Bluetooth Low Energy — a full 2.4GHz wireless protocol stack for short-range communication.	Phone pairing, local data sync, low-latency alerts when a phone is nearby.

Protocol	What it physically is	Used for
LTE-M/NB-IoT	Low-power cellular wireless standards, subsets of LTE optimized for small, infrequent IoT data.	Autonomous emergency alerting when BLE/phone is unavailable.
GNSS	Global Navigation Satellite System receiver (GPS and equivalents) — receives satellite signals to compute location.	Attaches a location fix to emergency alerts sent over LTE.

11. RF System

Radio-frequency (RF) design is often the least familiar area for a software/security engineer moving into hardware, so this section explains it carefully from the ground up.

11.1 Radio vs. Antenna — the Core Distinction

The "radio" lives inside a chip (nRF5340 for BLE, nRF9151 for LTE/GNSS) and is responsible for generating and interpreting an electrical high-frequency signal — modulating data onto a carrier wave and demodulating it back into data on receive. The "antenna" is a separate, passive physical structure whose only job is to convert that electrical signal into an electromagnetic wave that can travel through the air (transmit), and to do the reverse on receive. A radio without an antenna cannot communicate with anything; an antenna without a radio is just a piece of metal or ceramic.

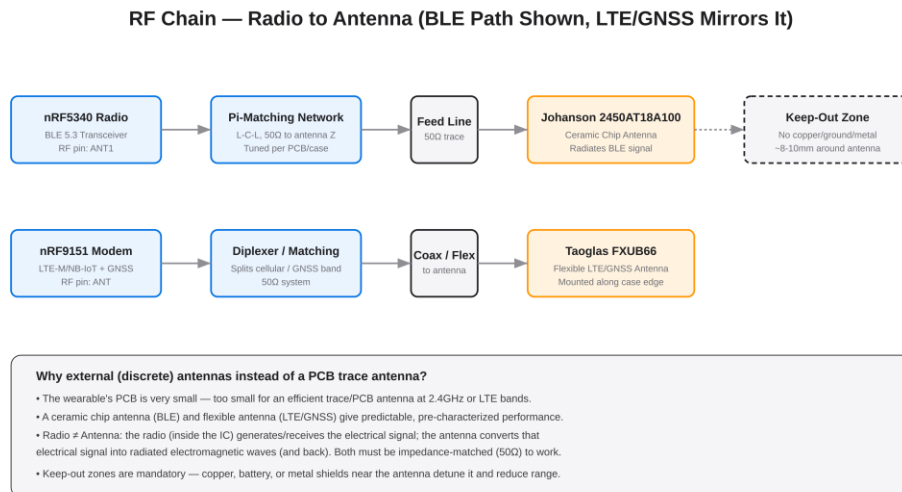


Figure 11.1 — RF chain from radio IC to antenna, both BLE and LTE/GNSS paths.

11.2 Why the Antennas Are External Components

- The PCB is far too small to act as an efficient trace antenna at 2.4GHz (BLE) or the lower LTE/GNSS frequencies without a much larger ground plane than a wristband allows.
- A pre-characterized, pre-certified discrete antenna (ceramic chip or flexible FPC) gives predictable performance the design team can rely on, rather than needing to design and validate a custom antenna structure from scratch.
- Discrete antennas can be positioned independently of the main PCB — important in a curved, space-constrained enclosure.

11.3 RF Matching Network

Every RF path needs to be "impedance matched" to 50 ohms — the standard reference impedance used throughout RF engineering — so that the maximum amount of signal power is transferred from the radio chip to the antenna, instead of being reflected back and wasted (or worse, causing damage). This is done with a small network of inductors

and capacitors (commonly a "Pi" or "L" network) placed between the radio's RF pin and the antenna feed. The exact values are only finalized after testing on the real PCB and enclosure, since the surrounding plastic, battery, and even the wearer's wrist all slightly shift the antenna's effective impedance.

11.4 Keep-Out Areas

A "keep-out area" is a zone around each antenna where no copper pour, ground plane, battery, or metal component is allowed. Any conductive material inside this zone absorbs or reflects part of the antenna's radiated field, reducing range and efficiency ("detuning"). The LTE/GNSS antenna requires the largest keep-out zone in this design, since it operates at lower frequencies (longer wavelengths) and is the device's life-safety communication path.

12. Supporting Components

Beyond the "headline" ICs and sensors, a real hardware design depends on dozens of small passive and support components most beginners overlook. This section explains what each type does and why it is non-negotiable.

Item	Why It Is Essential
32 MHz Crystal	A precision quartz timing reference for the nRF5340's CPU and radio. Without it, the radio cannot hit the tight frequency accuracy Bluetooth/cellular standards require.
32.768 kHz Crystal	A very low-power "watch crystal" used to keep time (real-time clock, sleep timers) while the rest of the chip is powered down — critical for multi-day battery life.
Decoupling Capacitors	Small capacitors (typically 100nF) placed right next to each IC power pin. They act as tiny local reservoirs that instantly supply current during fast digital switching, preventing voltage dips that could reset or glitch the chip.
Bulk Capacitors	Larger capacitors (1μF–100μF) that smooth out slower, larger current swings — such as the ~500mA burst the nRF9151 draws when transmitting.
Resistors	Used for current limiting, voltage division, and setting bias points throughout the analog and digital circuitry.
Pull-up Resistors	Required on every I2C bus and on GPIO inputs like the SOS button — they hold a digital line at a known "high" level when nothing is actively driving it low.
Inductors	Energy-storage elements essential to switching (buck) regulators, and part of RF matching networks.
Ferrite Beads	Small components that suppress high-frequency noise on power/signal lines without affecting the desired lower-frequency signal — useful for keeping RF or switching noise from spreading across the board.
RF Matching Components	The inductor/capacitor networks discussed in Section 11 that tune each radio's output impedance to its antenna.
ESD Protection	Small diodes placed on any pin exposed to the outside world (antenna feeds, the SOS button, the charging pins) that safely shunt static-discharge energy away from sensitive silicon.
Programming Header	A small connector (SWD/JTAG) used only during development and manufacturing test to flash firmware and debug the MCU — not present or needed in the final consumer-facing product beyond a possible pogo-pad test point.
Debug Connector	Same category as the programming header; sometimes a separate connector is used for the cellular modem's own debug/trace interface.
Ground Plane	A dedicated copper layer connected to circuit ground, spread across most of the PCB. It provides a low-impedance return path for all signals and shields sensitive circuitry from noise.
Power Plane	A dedicated copper layer (or large copper pour) distributing a given voltage rail with minimal resistance and inductance.
Copper Layers	The number of separate conductive layers inside the PCB stack-up (commonly 4 layers for a design this dense) — more layers allow more routing space and dedicated ground/power planes.
Test Points	Small exposed copper pads connected to key signals/rails, used during manufacturing test to verify the board is working correctly before final assembly.

13. Hardware Interconnection

The table below is the complete point-to-point connection list between major components — effectively a plain-language precursor to the schematic net list.

Component A	Component B	Protocol	Voltage	Purpose
BMI270	nRF5340	I2C	1.8–3.0V logic	Motion/fall data + interrupt
MAX86176	nRF5340	SPI	1.8–3.0V logic	PPG/ECG waveform streaming + interrupt
MAX30208	nRF5340	I2C	1.8–3.0V logic	Temperature reading + alert interrupt
IM69D130	nRF5340	PDM	1.8–3.3V logic	Digital audio stream
nRF5340	nRF9151	UART	1.8–3.0V logic	AT-command control, data payload for LTE uplink
nRF5340	BLE Antenna	RF (50Ω)	RF signal	Radiate/receive BLE 2.4GHz signal
nRF9151	LTE/GNSS Antenna	RF (50Ω)	RF signal	Radiate/receive cellular + GNSS signal
nPM1300	nRF5340	I2C	1.8–3.0V logic	Battery status, rail control, fault reporting
nPM1300	All ICs	Power rails	1.8V / 3.0V / 3.3V	Regulated power distribution
Battery	nPM1300	Direct wire	3.0–4.2V	Primary energy input
Pogo Connector	nPM1300	Direct wire	5V (charge input)	External charging input
SOS Button	nRF5340	GPIO	3.0–3.3V logic	Manual emergency trigger
nRF5340 / Haptic Driver IC	LRA Actuator	PWM/I2C + AC drive	Driver-specific	Haptic feedback pulses

14. Bill of Materials (BOM)

This is the engineering-level BOM covering major ICs and mechanical/electromechanical parts. Passive component quantities are itemized per-IC in Section 5's "Required External Components" tables and consolidated in the Section 19 shopping list.

Manufacturer	Part Number	Qty (per unit)	Function
Nordic Semiconductor	nRF5340-QKAA-R	1	Main MCU / Application Processor
Nordic Semiconductor	nRF9151-LACA-R	1	Cellular Connectivity (LTE-M / NB-IoT + GNSS)
Bosch Sensortec	BMI270	1	Motion Sensor (6-axis IMU)
Analog Devices (Maxim)	MAX86176	1	Health Sensor (PPG / ECG Analog Front-End)
Analog Devices (Maxim)	MAX30208	1	Body Temperature Sensor
Infineon Technologies	IM69D130	1	Audio Sensor (Digital MEMS Microphone)
Nordic Semiconductor	nPM1300	1	Power Management IC
Cell vendor (e.g., Lishen / EEMB / Grepow — customer-specified)	Custom 3.7V 250–300mAh flex pack	1	Energy Storage
Precision Microdrives	(e.g.) 8mm Linear Resonant Actuator, C-series	1	Haptic Feedback
Johanson Technology	2450AT18A100	1	RF — Bluetooth Antenna
Taoglas	FXUB66	1	RF — Cellular & GNSS Antenna
Alps Alpine	Low-profile waterproof SMD tactile switch (SKRG-series class)	1	User Input
Connector vendor (custom/standard 2-pin magnetic pogo)	2-Pin Magnetic Pogo Connector Set	1	Charging Interface

15. Mechanical Design

15.1 Target Dimensions

- Overall wearable "puck" size: approximately 42mm × 30mm × 11–13mm (length × width × thickness) — comparable to a premium fitness band module.
- Estimated thickness budget: ~1.5mm case wall + ~1.0mm PCB + ~2.5mm battery + ~1.5mm haptic/shielding + ~1.5mm optical gasket + case clearance ≈ 11-13mm total, consistent with premium wearable products on the market.
- Wristband strap: standard 20–22mm quick-release silicone band.

15.2 Key Mechanical Features

Feature	Design Consideration
Optical sensor window	Clear, scratch-resistant lens directly over MAX86176/MAX30208, sealed with a light-blocking gasket to prevent ambient-light interference with PPG readings.
Microphone acoustic port	Small waterproof mesh-covered opening aligned with the IM69D130, positioned away from the skin-contact face.
Charging pins	Gold-plated pogo pins recessed slightly from the case surface to prevent accidental shorting against metal objects, with a matching magnetic dock.
SOS button	Raised, textured button cap for tactile identification without looking, using a waterproof-sealed switch mechanism.
Antenna placement	BLE ceramic chip antenna at one PCB edge; LTE/GNSS flexible antenna routed along the opposite inner case wall, each within its keep-out zone.
Water resistance	Target IP67/IP68: fully gasket-sealed case halves (ultrasonic weld or adhesive), sealed optical window, waterproof SMD button, portless magnetic charging — no exposed connector to seal/fail.

16. Hardware Assembly Order

- 1. SMT (Surface-Mount Technology) placement and reflow soldering of all ICs and passives onto the bare main PCB.
- 2. Automated Optical Inspection (AOI) and In-Circuit Test (ICT) to catch solder defects before further assembly.
- 3. Flash initial bootloader/test firmware via the SWD programming header; run a functional self-test of every sensor and radio.
- 4. Solder or connect the battery pack to its PCB pads/connector; verify charge/discharge behavior through the nPM1300.
- 5. Attach the haptic actuator and any EMI shielding cans over the RF sections.
- 6. Install the optical light-block gasket over the PPG/temperature sensor area.
- 7. Seat the populated PCB assembly into the bottom case half, routing/folding the flexible LTE/GNSS antenna along its designated path.
- 8. Ultrasonically weld or adhesive-bond the top cover/lens to the bottom housing, forming the sealed enclosure.
- 9. Attach the wristband strap lugs.
- 10. Final end-of-line functional test, waterproofing (IP) test, and calibration (see Sections 17–18).

17. Manufacturing Considerations

17.1 PCB Layers

A 4-layer stack-up is recommended: Layer 1 (top) for component placement and signal routing, Layer 2 as a dedicated ground plane, Layer 3 as a power-distribution plane, and Layer 4 (bottom) for additional routing and any bottom-side components. This gives clean, low-impedance return paths for both digital and RF signals — essential given the density of this design.

17.2 Assembly

Standard SMT reflow assembly is appropriate for all components in this BOM, including the fine-pitch nRF5340/nRF9151 packages. The battery and any hand-soldered connectors are added in a secondary manual/semi-automated step after reflow, since lithium cells cannot go through a reflow oven.

17.3 Testing

Each board should pass Automated Optical Inspection (AOI), In-Circuit Test (ICT) for shorts/opens, and a functional test that exercises every sensor, both radios, the haptic driver, and the charging path before final enclosure assembly.

17.4 Calibration

The MAX86176 optical path benefits from a per-unit calibration step (measuring against a reference pulse oximeter) to compensate for small manufacturing variance in LED brightness and photodiode sensitivity. The BMI270 accelerometer/gyroscope should have factory offset calibration applied and stored in non-volatile memory.

18. Hardware Testing Plan

Component	Test Procedure
nRF5340	Flash test firmware over SWD; verify both cores boot, run a BLE advertising test, confirm all bus peripherals (I2C/SPI/UART/PDM) respond.
nRF9151	Verify UART AT-command link from the nRF5340; attach to a test cellular network; confirm GNSS cold-fix time and accuracy in open sky.
BMI270	Read and verify accelerometer/gyroscope values at rest match expected $1g/0^\circ/s$; verify interrupt fires under a controlled tap/motion test.
MAX86176	Verify LED drive current and photodiode ADC counts against a known reference; run a bench PPG capture and confirm a plausible heart-rate waveform.
MAX30208	Compare readings against a calibrated reference thermometer at multiple temperatures; verify ALERT interrupt fires at a programmed threshold.
IM69D130	Capture PDM audio and verify SNR/frequency response against a known test tone.
nPM1300	Verify each output rail voltage under load; run a full charge cycle and confirm CC/CV behavior and charge-complete reporting; verify fuel-gauge accuracy against a bench multimeter/coulomb counter.
LRA Haptic	Verify the haptic driver correctly auto-detects resonant frequency and produces a strong, repeatable vibration pulse.
BLE Antenna	Bench RF test: verify return loss (S11) and radiated power/range meet the Johanson reference spec.
LTE/GNSS Antenna	Bench RF test in the final enclosure: verify return loss across all target bands and GNSS sensitivity.
SOS Button	Mechanical actuation-cycle test; verify debounce firmware correctly registers single presses without false triggers.
Pogo Charging	Verify charge current/voltage under normal and reverse-polarity/misalignment conditions (protection circuit test).

19. Final Hardware Summary

This design combines a dual-core Nordic nRF5340 as the central compute and Bluetooth hub, a Nordic nRF9151 for autonomous LTE-M/NB-IoT emergency communication with GNSS positioning, and a focused set of best-in-class sensors — Bosch BMI270 for motion/fall detection, Analog Devices MAX86176 for cardiac/PPG monitoring, MAX30208 for precision temperature, and Infineon IM69D130 for acoustic event detection. A Nordic nPM1300 PMIC unifies battery charging, voltage regulation, and fuel-gauging in a single chip, powering the whole system from a compact 250–300mAh Li-Po cell.

The architecture's defining safety principle — a manual SOS button and an independent cellular uplink, so the device never relies solely on AI judgment or a paired phone — is reflected throughout every section of this document, from the signal-flow branching logic in Section 8 to the RF keep-out priorities in Section 11.

This document is intended as the starting reference for schematic capture in KiCad or Altium Designer. All component values are reference-design starting points (Section 5) and must be verified against the latest datasheet revisions; all RF matching networks (Section 11) must be tuned on real hardware before production.

Purchasing Checklist — Complete Prototype Shopping List

Every physical part required to build one working prototype, organized by category. Quantities are per single unit; multiply for prototype batch size.

A. Major ICs / Modules

Part	Manufacturer	Qty
nRF5340 SoC	Nordic Semiconductor	1
nRF9151 SiP	Nordic Semiconductor	1
BMI270 IMU	Bosch Sensortec	1
MAX86176 Health AFE	Analog Devices (Maxim)	1
MAX30208 Temp Sensor	Analog Devices (Maxim)	1
IM69D130 MEMS Microphone	Infineon Technologies	1
nPM1300 PMIC	Nordic Semiconductor	1

B. Electromechanical / RF Parts

Part	Manufacturer	Qty
Li-Po Battery Pack	Cell vendor (e.g., Lishen / EEMB / Grepow — customer-specified)	1
LRA Haptic Actuator	Precision Microdrives	1
BLE Ceramic Chip Antenna	Johanson Technology	1
LTE/GNSS Flexible Antenna	Taoglas	1
SOS Emergency Button	Alps Alpine	1
Magnetic Pogo Pin Connector	Connector vendor (custom/standard 2-pin magnetic pogo)	1

C. Crystals & Oscillators

Part	Value	Qty
HFXO Crystal	32 MHz, CL 9-12pF	1
LFXO Crystal	32.768 kHz, CL 9-12.5pF	1

D. Passives (Capacitors, Resistors, Inductors)

Aggregate estimate across all component reference designs (Section 5); exact final counts are confirmed at schematic capture:

Category	Typical Value Range	Approx. Qty (whole board)
Decoupling capacitors	100nF, 0201/0402	~25-30
Bulk/reservoir capacitors	1 μ F–100 μ F	~10-12
Crystal load capacitors	9-12pF	4
Pull-up resistors	2.2k Ω –10k Ω	~8-10
RF matching inductors/capacitors	0402, per antenna app notes	~10 (both antennas)
Buck regulator inductors	1.0–2.2 μ H shielded	2
ESD protection diodes	Low-Cap TVS, 0402	~5

E. Connectors & Mechanical Hardware

Part	Qty
SWD/JTAG programming header (2×5, 1.27mm)	1
U.FL/MHF coax pigtail (if not direct-solder)	1
Nano-SIM socket or eSIM chip	1
Battery connector (JST-SH 1.0mm, if not direct-solder)	1
Case top cover / lens	1
Case bottom housing	1
Optical light-block gasket	1
Acoustic mesh (microphone port)	1
Wristband strap + quick-release lugs	1 set
Haptic driver IC (e.g., DRV2605L-class)	1

F. PCB

Item	Spec
Bare PCB	4-layer, ENIG finish, controlled-impedance RF traces
PCB thickness	Typically 0.8–1.0mm for this stack-up

Note: This checklist covers one working prototype at the electronics/mechanical level. It does not include production-only items (stencils, fixtures, injection-mold tooling) which are only needed once the design moves from prototype to mass production.